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✓





Next >

6.1.3 What you will learn

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- Recognize how floating point numbers are stored.
- Employ strategies for avoiding unnecessary overflow and underflow that can occur in intermediate computations.
- Compute the machine epsilon (also called the unit roundoff) for a given floating point representation.
- Quantify errors in storing real numbers as floating point numbers and bound the incurred relative error in terms of the machine epsilon.
- Analyze error incurred in floating point computation using the Standard Computation Model (SCM) and the Alternative Computation Model (ACM) to determine their forward and backward results.
- Distinguish between conditioning of a problem and stability of an algorithm.
- Derive error results for simple linear algebra computations.
- State and interpret error results for solving linear systems.
- Argue how backward error can affect the relative error in the solution of a linear system.

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